an exchange modifying its systems to implement the Tick Size Pilot, which, based upon the input from commenters, the Commission estimated to be around $140,000.[1901] The Commission also estimated in the Proposing Release that the implementation cost for a broker-dealer to update its systems, including its SOR, would be $9,000.[1902] One commenter stated that the Commission significantly underestimated the costs for a trading venue to update its systems and estimated that its costs to modify its trading venue to account for the changes in round lot size and order protection would be between $3.4 and $4 million.[1903] The Commission agrees and believes that the estimates from the Proposing Release underestimated the implementation costs for modifying trading venue and broker-dealer systems to account for the new round lot definition and changes in the Order Protection Rule, which created a separate NBBO and PBBO.[1904] However, the Commission also believes, as suggested by commenters, that the new round lot definition under the final amendments will require significantly less system modifications compared to the Proposing Release.[1905] For example, one commenter stated that if the new round lot definitions were protected then trading venues and broker-dealers will be able to rely on existing technology

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1901 In the Proposing Release, the Commission stated that it did not have detailed information on the operation of exchange matching engines and believed that the $140,000 from the Tick Size Pilot may provide some sense of the level of cost associated with the changes SROs, ATSs, and other off-exchange trading venues would have to make. See Proposing Release, 85 FR at Section VI.C.1(c)(i).

1902 See id.

1903 See Nasdaq Letter IV at 17. This commenter also estimated it would cost an additional $800,000 to $1.2 million to modify its systems to account for the changes in locked and crossed markets as a result of the changes in order protection. See Nasdaq Letter IV at 19.

1904 See supra Section II.E.1.

1905 See, e.g.; MEMX Letter at 4; BestEx Research Letter at 6-9.

to continue to operate without significant changes to current execution and routing logic compared to having to build new logic and workflow to account for a separate NBBO and PBBO.[1906] Additionally the Commission believes that many broker-dealer and trading venue systems already account for different round lot sizes and will not need to make extensive modifications to account for a changing round lot size each month. Therefore, although the implementation costs estimated in the Proposal Release may have underestimated the costs to modify systems to account for a separate NBBO and PBBO, the Commission believes they provide an appropriate sense of the level of cost associated with the implementation costs of modifying systems related to the new round lot definition under the final amendments, including building or modifying systems to account for the monthly change in a securities round lot size. The Commission estimates that a trading venue will incur an initial implementation cost of approximately $140,000 and a broker-dealer will incur an initial implementation cost of approximately $9,000 to modify its systems to account for the new round lot definition. However, these costs will vary substantially according to the existing infrastructure of the broker-dealer or trading venue.

(vii) Other Rules and Regulations

The amendments to the definition of round lot and resulting mechanical changes to the NBBO spread, affect how other rules and regulations operate.[1907] In particular, this change

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1906 See MEMX Letter at 4.

1907 The Commission is also deleting the reference to "The Nasdaq Stock Market, Inc." from the definition of protected bid or offer and believes that this change will have no economic effects. As explained above, Nasdaq is now a national securities exchange and is thus otherwise bound by the definition. See supra note 361.

affects which orders determine the reference price for numerous rules, including rules under the Exchange Act, SRO rules, and effective national market system plans.[1908] Specifically, the Commission believes that the changes to the NBBO may present changes to the benchmark prices used in Regulation SHO, LULD, retail liquidity programs, market maker obligations, and certain exchange order types and recognizes that the change in the benchmark price may result in economic effects. Further, changing the NBBO will alter the estimation mechanics for Rule 605 metrics, resulting in implementation costs. In addition, the round lot definition will result in economic effects through its impact on the 17 CFR 242.606 (Rule 606) compliance. Finally, although the new round lot definition may alter the requirements of some rules, such as Rules 602, 604, and 610(c), the Commission believes that the economic effects of the changes are uncertain and depend on current practices of handling odd-lot-sized orders. If broker-dealers already include odd-lot-sized orders when complying with the provisions of these rules, then the new round lot definition may not produce any economic effects related to these rules.

For the Short Sale Circuit Breaker, the reference bid for the execution of a short sale transaction could be higher for stocks in the higher priced round lot tiers under the final amendments than it is currently, potentially slightly increasing the burdens on short selling.[1909] Currently, after the Short Sale Circuit Breaker triggers, short sales can only execute at prices greater than the national best bid. While short sales are currently permitted to execute against any odd-lot quotations that exist above the national best bid, the new round lot definition will

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[1908] The Commission discussed many of these changes in the Proposing Release. See Proposing Release, 85 FR at Section III.C.1(d)(i).

[1909] One commenter stated that the Commission failed to include analysis of how the change in the round lot definition affected Rule 201 of Regulation SHO. See NYSE Letter II at 8. This commenter is mistaken. The Commission did qualitatively analyze the effects of the round lot definition on Rule 201 of Regulation SHO. See Proposing Release, 85 FR at Section VI.C.1(c)(iii).

reduce the instances of such odd-lot quotations in higher priced stocks. Therefore, the final amendments may result in a higher national best bid and thus result in a slightly higher benchmark price for short sale executions in stocks priced more than $250, reducing the fill rate of short sales or increasing the time to fill for short sales.

In addition, a potentially higher national best bid (or lower national best offer) price could marginally affect the trigger of the Short Sale Circuit Breaker. In particular, the final amendments could result in slight delays in or a reduction in the number of Short Sale Circuit Breaker triggers, or it could have the opposite effect in the nine stocks whose round lot size will increase. In particular, a national best bid that includes smaller round lots could result in a higher-priced execution relative to a national best bid that does not include smaller round lots. This higher-priced execution could be above the price that would trigger the Short Sale Circuit Breaker whereas an execution on a 100-share quote would have triggered the circuit breaker. This could delay the trigger if the price continues downward, such that the circuit breaker still triggers, or the circuit breaker may not trigger at all if the price rebounds after such an execution. On the other hand, in the eight stocks that will have a larger round lot size, and lower priced national best bid, it could have the opposite effect on circuit breaker triggers: triggering sooner and more often.[1910]

The Commission believes that the economic effects of the potential impact on the Short Sale Circuit Breaker are unlikely to be significant. These effects should not create implementation costs, and the Short Sale Circuit Breaker should continue to function consistent with its stated purpose. Notably, if the adopted rule will result in not triggering as many Short

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[1910] See supra Section V.C.1(b)(i).

Sale Circuit Breakers, it could reduce ongoing compliance costs in situations in which the price rebounds despite the lack of a price test on short sales.

Similarly, a potentially higher bid price or lower offer price could affect the trigger of a Limit State under the LULD Plan. A lower-priced national best offer or a high-priced national best bid could result in that quote being more likely to touch a price band, thus triggering a Limit State, when it otherwise would not have. Depending on whether the quote would have otherwise rebounded, this could increase the number of Limit States and/or Trading Pauses or could merely trigger such Limit States or Trading pauses sooner. As in the case of the Short Sale Circuit Breaker, the effects should not create implementation costs, and LULD should continue to function consistent with its stated purposes. In addition, the economic effects of this potential marginal change depends largely on how often odd-lot quotations lead price declines or lead price increases.

As discussed in the Proposing Release,[1911] a number of Rule 605 execution quality statistics are benchmarked to the NBBO. Under the final amendments, the NBBO will be based on the tiered, price-based round lot sizes, which means any Rule 605 execution quality statistics that rely on the NBBO as a benchmark will reflect the modified definition of the NBBO. This could cause certain execution quality statistics to change in higher priced stocks. As discussed above, the Commission believes that the NBBO will become narrower for some stocks in higher price tiers. This could cause execution quality statistics that are measured against the NBBO to change because they will be measured against the new, narrower NBBO. For example,

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[1911] See Proposing Release, 85 FR at Section III.C.1(d)(i).

execution quality statistics on price improvement for higher priced stocks may show a reduction in the number of shares of marketable orders that received price improvement because price improvement will be measured against a narrower NBBO.[1912] However, the Commission believes that some of these changes may cause some Rule 605 statistics to more accurately reflect actual execution quality because the NBBO based on the new definition for round lots may now take into account more liquidity that the current NBBO ignores.[1913] The Commission believes that these effects will be larger for stocks in higher price tiers because their new round lot definition will include fewer shares.

In addition, the NBBO midpoint in stocks priced higher than $250 could be different under the adopted rules than it otherwise would be, resulting in changes in the estimates for Rule 605 statistics calculated using the NBBO midpoint, such as effective spreads. In particular, at times when bid odd-lot quotations exist within the current NBBO but no odd-lot offer quotations exist (and vice versa), the midpoint of the NBBO resulting from the rule will be higher than the current NBBO midpoint. For example, if the NBB is $260 and the national best offer is $260.10,

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1912 A commenter agreed that the smaller round lot size would cause a decrease in the number of orders showing price improvement in Rule 605 statistics. See Nasdaq Letter IV at 19.

1913 In the hypothetical case of a stock in which there are often valuable odd-lot quotes, broker-dealers trading in this stock can currently use these odd-lot quotes to improve on the NBBO, and this improvement might be reflected in Rule 605 statistics. Under the new round lot definition, if this stock is priced over $250 per share, then some of these odd-lot quotes could end up being defined as round lots under the new definition and thereby end up the basis for the NBBO. With these quotes as the NBBO, the broker-dealer will no longer appear to be improving over the NBBO in its execution, and Rule 605 statistics may appear to indicate a decrease in execution quality. However, they will, in fact, merely be reflecting a more accurate picture of the market circumstances at the time of execution. One commenter agreed that Rule 605 statistics may not be accurate because they do not include information on odd-lot quotes priced better than the NBBO. See Healthy Markets Letter at 15. One commenter agreed that the new round lot size would improve the accuracy of Rule 605 statistics and that this would improve transaction cost analysis for funds that rely upon these statistics to analyze broker-dealer execution quality. See ICI Comment Letter at 7.

the NBBO midpoint is $260.05. Under the adopted rules a 40 share buy quotation at $260.02 will increase the NBBO midpoint to $260.06. Using this new midpoint, effective spread calculations for buy orders will be lower but will be higher for sell orders. More broadly, the adopted rules will have these effects whenever the new round lot bids do not exactly balance the new round lot offers. However the Commission does not know to what extent or direction that odd-lot imbalances in higher priced stocks currently exist, so it is uncertain of the extent or direction of the change.

Finally, the Commission recognizes that the new round lot definitions could force market centers (or their third-party service providers) to revise their processes for estimating the Rule 605 execution statistics. Such changes will result in implementation costs.

The Commission recognizes that the NBBO serves as a benchmark in SRO rules in addition to Exchange Act rules and effective national market system plans. For example, the NBBO acts as a benchmark for various retail liquidity programs on exchanges, for exchange market maker obligations, for some order types, and for potentially many other purposes.[1914] As such, including smaller quotes in the NBBO will change how these rules operate and these changes could have economic effects. For example, having to post more aggressive limit orders into retail liquidity programs could reduce the already low volume by reducing the liquidity available but could result in better prices for those retail investors able to execute against that liquidity. In addition, a narrower NBBO could effectively increase some market maker obligations, which could improve execution quality for investors and/or provide a disincentive to being a market maker on the margin. Alternatively, the exchanges with such retail liquidity

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[1914] See supra Section V.C.1(b)(i) for a discussion of the effect of changes to the NBBO on order types and for a discussion related to changes to round lot size for stocks with round lots of less than 100 shares.

programs, order types, or market maker obligations could elect to propose rule changes to maintain the current operation of these rules. Such proposals could mitigate any follow-on economic effects (both benefits and costs) but would require exchanges to incur the expenses associated with proposing amendments to their rules.

As discussed in the Proposing Release,[1915] the definition of round lot could result in an increase in the number of indications of interest in higher priced stocks that will be required to be included in 606(b)(3) reports. Depending on the number of potential indications of interest included as a result of the final amendments, the Commission believes that these changes could increase the benefits of 17 CFR 242.606(b)(3) (Rule 606(b)(3)) with little to no effect on costs.[1916] In particular, the inclusion could result in clients receiving information on order routing for more of their orders, with the resulting benefits. Further, because the incremental cost of adding orders to the reports is low, the Commission does not expect that adding indications of interest to the reports will significantly increase costs.

One commenter stated that the Commission did not examine the effects of the new rules on Rule 603(b), the Vendor Display Rule.[1917] The new round lot definition will require broker-dealers to show a consolidated display that includes the NBBO derived from the new round lot size in higher priced stocks. This will allow investors to see odd-lot quote information that may not previously have been included in the NBBO under the current round lot definition, which

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1915 See Proposing Release, 85 FR at Section III.C.1. for a discussion of how the definition impacts Rule 606.

1916 See Proposing Release, 85 FR at n.227 for a discussion of the benefits of 606(b)(3).

1917 See NYSE Letter II at 6–7.

may improve their trading decisions and order routing and execution quality.[1918] Broker-dealers may also incur implementation costs in order to adjust their systems.[1919]

The new round lot definition would also affect the requirements regarding the size of orders that need to be collected and made available under Rules 602(a), 602(b), 604(a)(1), and 604(a)(2). However, it is unclear whether this will have any economic effects, because it would depend on the current practices for handling odd-lot orders. For example, exchanges may already have procedures to collect and make available their best bids and offers to vendors, regardless of the size of those best bids and offers. Further, broker-dealers may already treat all bids and offers as firm quotes regardless of size and may already display all customer limit orders regardless of size. To the extent that these practices are in place, there will be no economic effect from these changes. To the extent that these practices are not in place, the final amendments will increase transparency in higher priced stocks by requiring broker-dealers and trading venues to include smaller sized orders that meet the new round lot definition under these rules.[1920] Broker-dealers and trading venues may also incur implementation costs in order to adjust their systems.[1921]

One commenter stated that the Commission did not consider the burden that applying Rule 610(c) to the new round lot definition would have on market participants and competition,

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1918 See supra Section V.C.1(b)(ii).

1919 See supra Section V.C.1(b)(vi)

1920 See supra Section V.C.1(b)(ii)

1921 One commenter stated that market makers would need to make adjustments to their systems to display customer limit orders in the new round lot sizes under Rule 604. See MFA Letter at 12–13. These costs are included in the costs to adjust systems to the new round lot size. See supra Section V.C.1(b)(vi)

including trading centers that display quotes.[1922] The Commission does not believe that applying the new round lot definition to Rule 610(c) create a significant burden for market participants, including trading centers that display quotes, or have a significant impact on competition. The Commission believes that exchanges may already pay the same rebates or charge the same access fees regardless of order size. Therefore, it does not expect the new round lot definition to affect these fees.

(c) Expanded Core Data Content

The Commission is adopting amendments to include certain information on odd-lot quotes at and inside the NBBO, certain depth of book data, and information on orders participating in auctions in the definition of core data. This section discusses the economic effects of expanding the core data content separately for each additional core data element and then discusses the additional costs that may accrue to market participants from the combined new core data elements, although competing consolidators will not be required to offer consolidated market products that include all of the content of expanded core data and market participants may choose not to take in all of the new core data elements in every instance.[1923] The economic effects discussed in this section depend on the fees for data content underlying core data charged by the effective national market system plan(s) for NMS stocks and the competing consolidators. The fees for data content underlying new core data are discussed later, in Section V.C.2(b).

The Commission believes that expanding the content of core data to include information on odd-lot quotes at and inside the NBBO, depth of book information, and auction information

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1922 See NYSE Letter II at 7–8.

1923 See infra Section V.C.1(c)(iv).